



MG-1S农业植保机拆装机讲解

Disassembling and Assembling the MG-1S

飞行器拆卸

Disassembling the Aircraft

注意事项：

- 1、壳类螺丝（上壳、下壳、后壳）主要都是内六角2.0，上下碳板螺丝主要是内六角2.5；
- 2、拆装顺序：上壳、上碳板→各功能模块（GPS、主控、PMU、中心板、断电续航）→下壳→机臂→电机电调→小臂大臂三通；

组装顺序：小臂大臂三通（注意黄白电调线）→电机电调（注意电机线的焊接顺序）→机臂（注意4个机臂的顺序）→下壳（注意RF线、中心板CAN线）→脚架支撑柱+碳板连接件→各功能模块（注意整理线材）→上碳板+后壳→左右脚架→上壳

- 3、焊接时注意焊点需光滑饱满，附近无锡渣。

Noticed：

- 1、The screw for plastic parts(top shell, bottom shell, back cover) are mainly 2.0 hexagonal socket screw, for upper and lower carbon board are mainly 2.5 hexagonal socket screw.
- 2、Disassembling order: top shell, carbon board →inner modules (GPS, main controller, PMU, central board, reserved battery module) →bottom shell →arms →motor and ESC →big arm, small arm and Y-Tee part.
- 3、Assembling order: big arm, small arm and Y-Tee part (pay attention to the yellow and white ESC cable) →motor and ESC (pay attention to the soldering order) →arm (pay attention to the order of four arms) →bottom shell (pay attention to the RF cable and central board CAN cable) →landing gear support +carbon board connector →inner modules (arrange cables) →upper carbon board +back cover →landing gears →top cover
- 3、Make sure there are no solder splash around, and the soldering spot should be full and smooth.



使用2.5内六角螺丝刀拧出固定电源信号插头的3颗螺丝，取出电源信号插头；
使用相同的方式取下固定在左脚架的电源插头；

1. Unscrew the three screws on the power signal adapter with a 2.5 hexagonal socket screwdriver.
2. Remove the power signal adapter.
3. Remove the power adapter on the left landing gear in the same way.



使用2.5内六角螺丝刀拧出右脚架支撑柱上的3颗螺丝，取出防脱片；
缓慢抽出右脚架，分离指南针；
使用相同的方法取出左脚架；

1. Unscrew the three screws on the right landing gear fixing piece with a 2.5 hexagonal socket screwdriver. Remove the anti-drop pieces.
2. Gently take out the right landing gear. Do not damage the compass cable.
3. Remove the left landing gear in the same way.



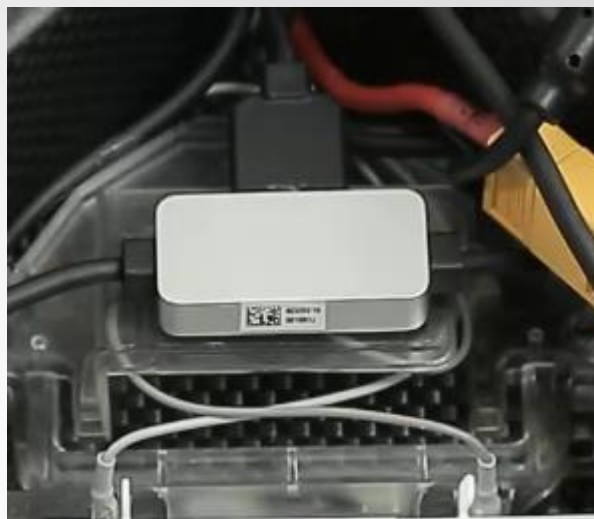
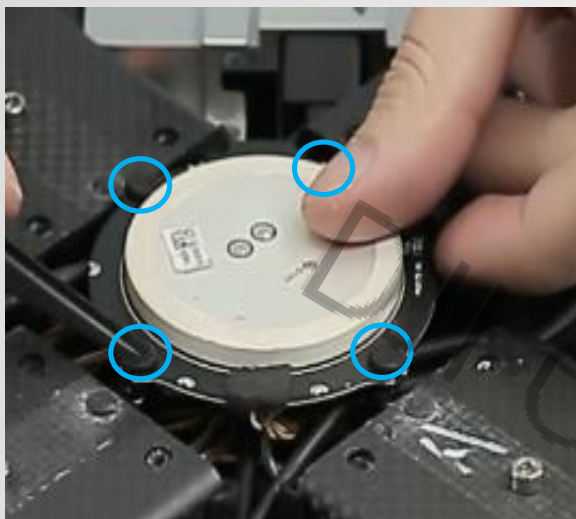
使用3.0内六角螺丝刀拧出固定脚架支撑柱的2颗螺丝，依次取出左右脚架支撑柱；

Unscrew the two screws on the landing gear fixing pieces with a 3.0 hexagonal socket screwdriver. Remove the left landing gear fixing piece and right landing gear fixing piece.



使用2.0内六角螺丝刀拧出固定上壳的6颗螺丝，取下上壳；
拧出固定上碳的螺丝，取出上碳板。

1. Unscrew the six screws on the upper aircraft shell with a 2.0 hexagonal socket screwdriver. Remove the upper shell.
2. Unscrew the screws on the upper carbon board. Remove the upper carbon board.



使用镊子挑开橡胶柱，取出GPS模块；

使用剪钳剪掉线材扎带；

取下粘贴在塑胶支架上GPS拓展CAN口，拔出连接在CAN口上气压计连接线；

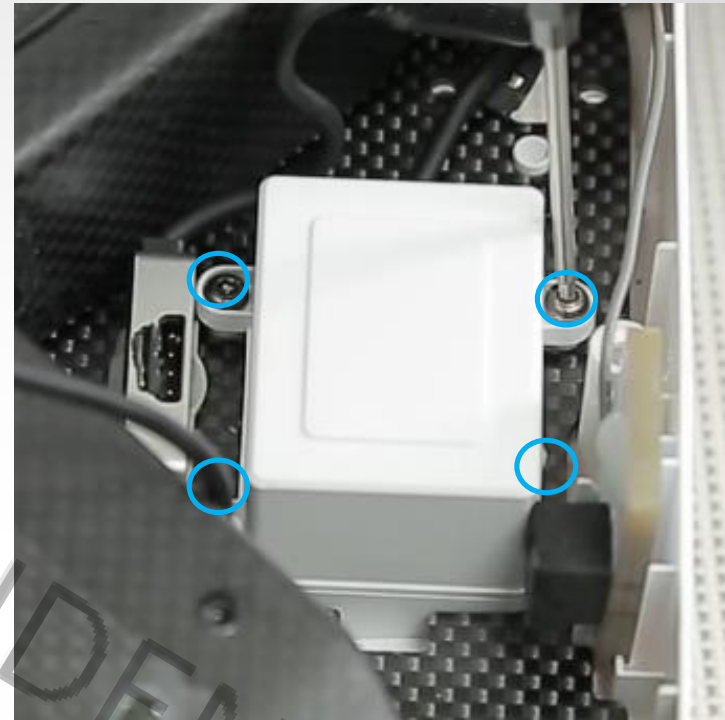
断开连接在主控上CAN1接口的GPS连接线，取出GPS模块；

1. Remove the rubber mounting piece with tweezers and remove the GPS module.
2. Cut off the cable ties with cutting pliers. Be careful not to damage the cables.
3. Remove the GPS CAN port on the bracket. Unplug the barometer cable on the CAN port.
4. Unplug the GPS cable on the CAN 1 port. Remove the GPS module.



拧出固定压块的1颗螺丝，先后取出固定主控压块和主控；
依次断开连接在主控上的RTK转接信号线，断电续航CAN线，422信号线，PMU电源线，LED灯线，
天空端信号线，电调线，断电续航信号线

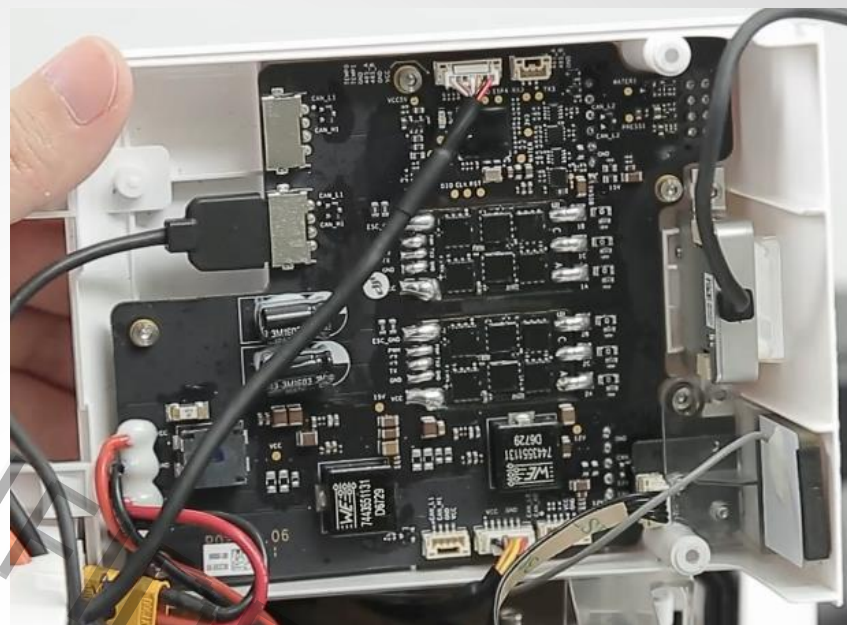
1. Unscrew the screw on the mounting piece. Remove the main controller mounting piece and main controller.
2. Unplug the RTK adapter cable, reserved battery module CAN cable, 422 signal cable, PMU power cable, LED cable, air system signal cable, ESC cables, and reserved battery module signal cable.



断开PMU与智能插头模组的电源连接线，取出PMU；

使用2.0螺丝刀拧下固定小白壳的4颗螺丝，取下小白壳；

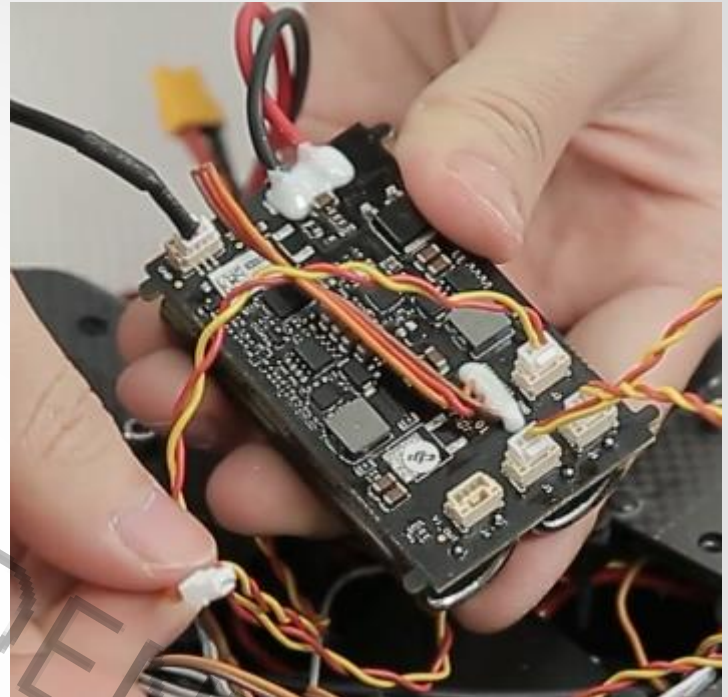
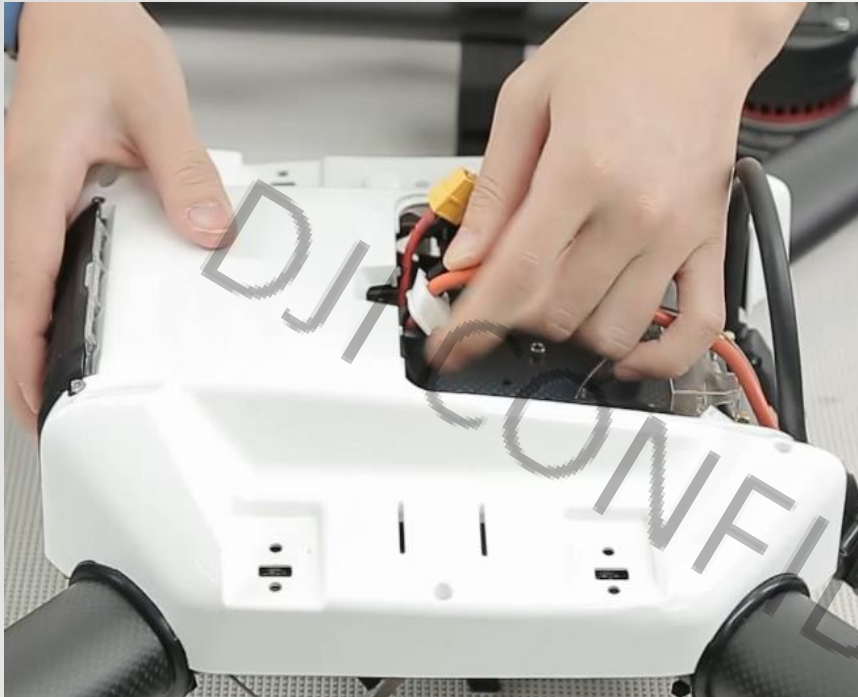
1. Unplug the power cable between the PMU and intelligent adapter module. Remove the PMU module.
2. Unscrew four screws on the white box, remove the white box;



使用2.0内六角螺丝刀拧出固定后壳的2颗螺丝；

依次拔出中心板上的智能插头信号线，中心板通讯线，RTK转接线，天空端对频转接线、天线、中心板电源线和断电续航电源线，取下中心板；

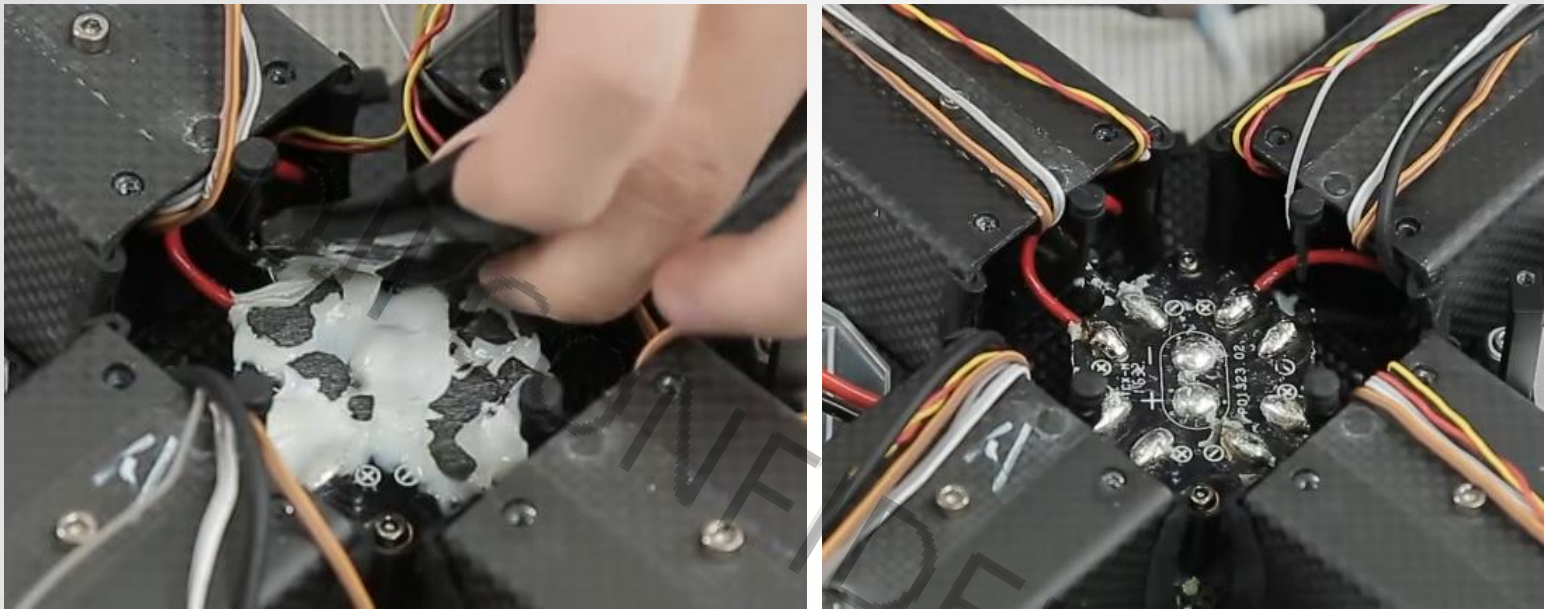
1. Unscrew the two screws on the rear shell with a 2.0 hexagonal socket screwdriver.
2. Unplug the intelligent adapter signal cable, central board communication cable, RTK adapter cable, and air system linking adapter cable, central board power cable and reserved battery module power cable. Remove the central board.



分离电源总线，取下下壳组件；

依次拔出连接在断电续航模块上4根电调馈线，取下断电续航模块

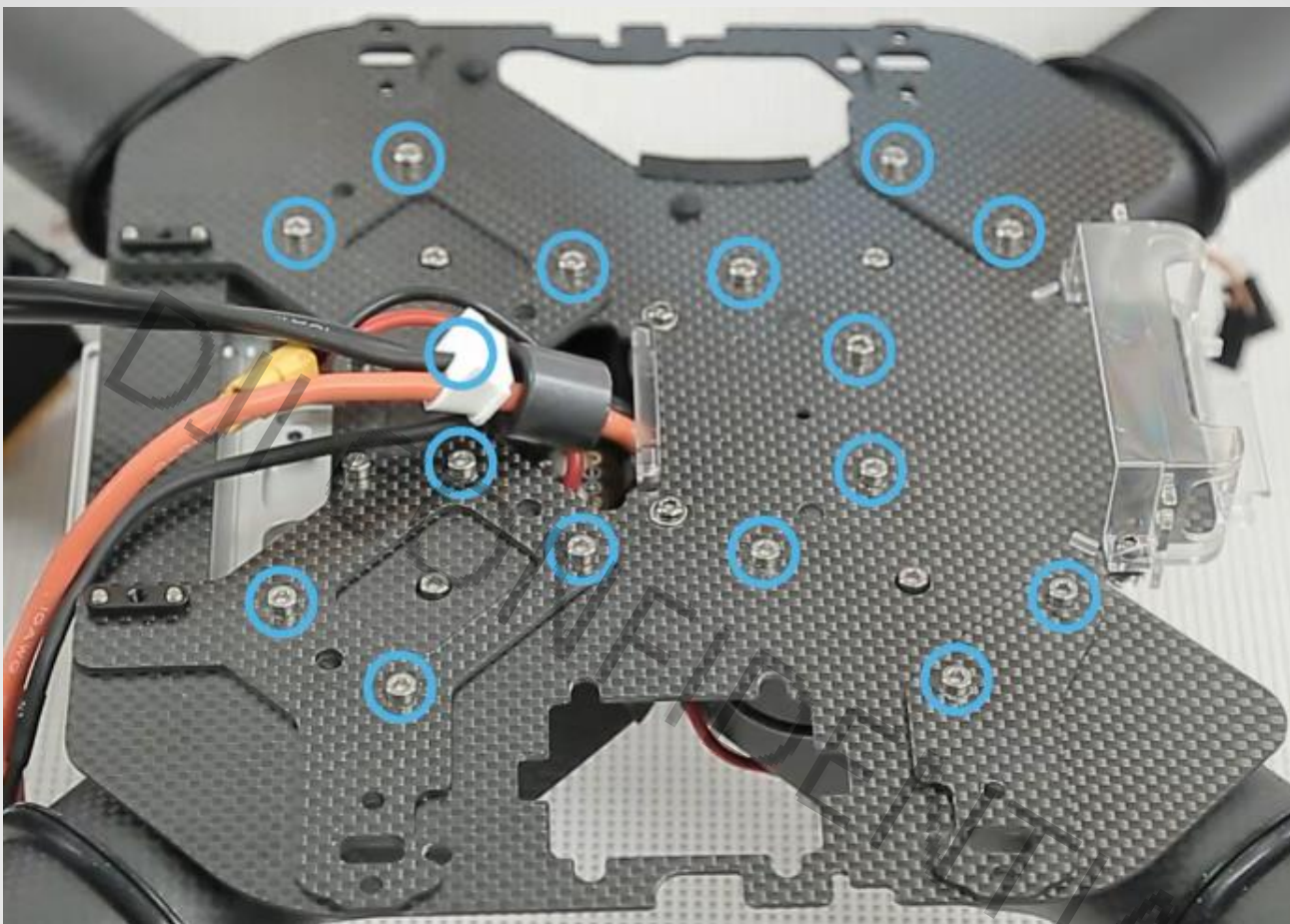
1. Remove the power cable and rear shell module.
2. Unplug the four feeders on the reserved battery module. Remove the reserved battery module.



撕下分电板上的黑色泡棉，使用镊子去除上面的白胶；

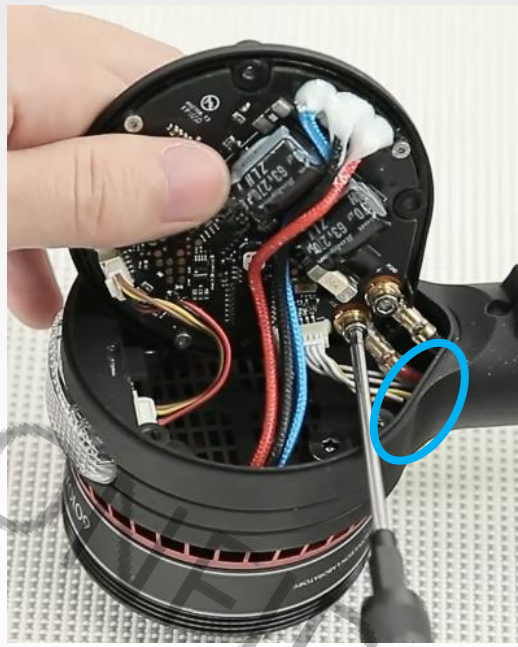
设置烙铁温度为 410°C - 430°C ，依次焊下分电板上的电源线；

1. Tear off the black foam on the power distribution board.
2. Set the temperature range of the soldering iron from 410°C to 430°C and disconnect the power cables on the power distribution board separately.



翻转至另一面，使用2.5内六角螺丝刀拧出固定下碳板的16颗螺丝，分离下碳板和机臂，注意分离机臂请标记1号、2号、3号、4号机臂，避免组装时混淆

Turn the aircraft right-side up. Unscrew the 16 screws on the lower carbon board with a 2.5 hexagonal socket screwdriver. Separate the lower carbon board and the arms. Mark arm 1, arm 2, arm 3, and arm 4.



使用2.5内六角螺丝刀拧出固定电机底盖上的4颗螺丝，取出电机底盖；
使用2.5内六角螺丝刀拧出固定电源线上的2颗螺丝，取出螺丝和垫片；
取出连接在电调上的信号线；

1. Use a 2.5 hexagonal socket screwdriver to remove the four screws on the bottom motor cover and detach the cover.
2. Use a 2.5 hexagonal socket screwdriver to remove the two screws for the power cable and detach the gasket.
3. Remove the signal cable on the ESC board.



拧下固定电机座上底部的2颗螺丝和固定电机座上面的1颗螺丝，分离电机座和碳管；
使用镊子去除电调上的白胶，将烙铁的温度设置为 410°C - 430°C ，焊下三根电机线，取下电调。

1. Remove the two screws on the bottom motor base and the screw on the motor base. Separate the base and carbon tube.
2. Clean the white glue on the ESC with tweezers. Set the temperature range of the soldering iron from 410°C to 430°C and remove three motor cables. Remove the ESC.

装机流程

Assembling the MG-1S



将天线沿三通主体孔穿出；

并将天线橡胶套穿过天线安装至三通上，整理天线；

1. Pass the antenna through the hole on the Y-tee part.
2. Attach the antenna rubber to the Y-tee part. Arrange the antenna.



将另一个天线从三通另一面孔穿出，并安装在三通上；
将天线帽安装至三通上，并锁上1颗螺丝固定

1. Pass the antenna through the hole on the Y-tee part and attach the antenna to the part.
2. Mount the antenna cover to the Y-tee part and tighten it with a screw.



将Y型复合线两端分别沿三通主体侧孔穿出；

注意Y型复合线一边为CW电调线，另一边为CCW电调线；

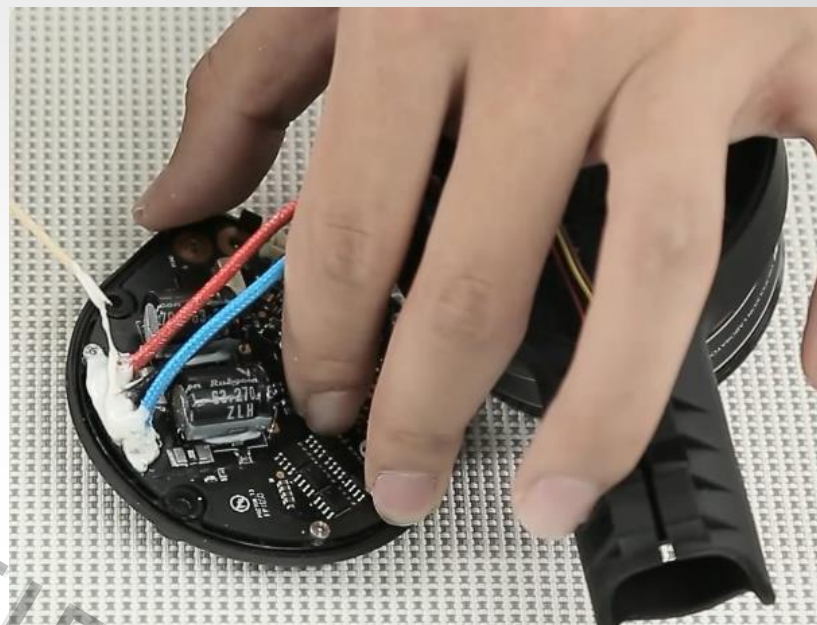
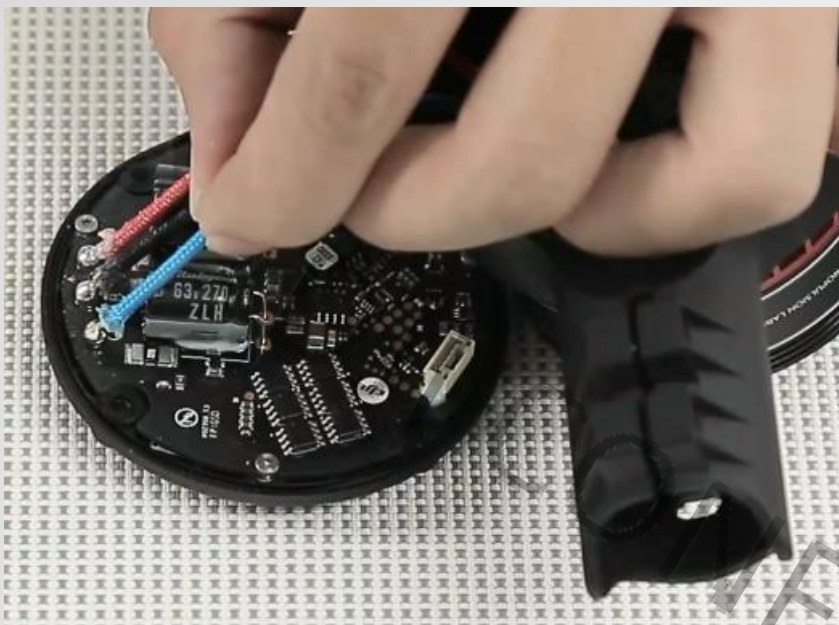
将机臂碳管上至对准碳管上螺丝孔，注意碳管带橡胶柱的一端为上方，并锁上4颗螺丝固定

1. Pass one end of the Y-type composite cable through the hole on the Y-tee part. Note that one end of the Y-type composite cable is the CW ESC cable and the other end the CCW ESC cable.
2. Align with the holes and mount the Y-tee part to the carbon tube. Make sure the carbon tube with rubber piece is on the top. Then secure the four screws.



将Y型复合线一段线材沿小臂孔穿出，并锁上1颗螺丝将小臂和三通主体固定；
使用相同方式安装另一侧的小臂

1. Pass one end of the Y-type composite cable through the hole on the small arm. Tighten the screw to fix the small arm and Y-tee part.
2. Mount the other small arm in the same way.



将电机线焊接在电调上，CCW电机红线对应A，黑线对应B，蓝线对应C；

CW电机红线对应A，蓝线对应B，黑线对应C；

焊接后使用洗板水清洁焊点，并刷适量的绝缘漆，并涂上白胶，注意白胶不能溢出电调板之外；

1. Weld the power cables onto the ESC board. The red cable on the CCW motor should be welded onto the spot A, black cable onto spot B, and blue cable onto spot C. The red cable on the CW motor should be welded onto the spot A, blue cable onto spot B, and black cable onto spot C.
2. Clean the welding spots and apply the welding part with insulation paint and white glue. Be careful not to let the glue spill over to the ESC board.



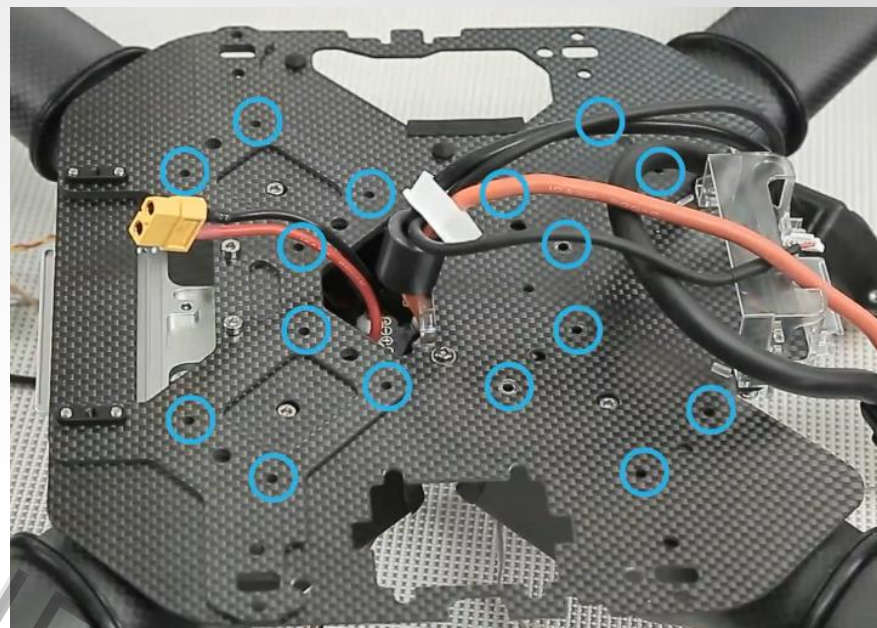
将电机电调安装在小臂，并拧紧固定电机底座的3颗螺丝；

Mount the motor and ESC to the small arm. Tighten the motor base with three screws.



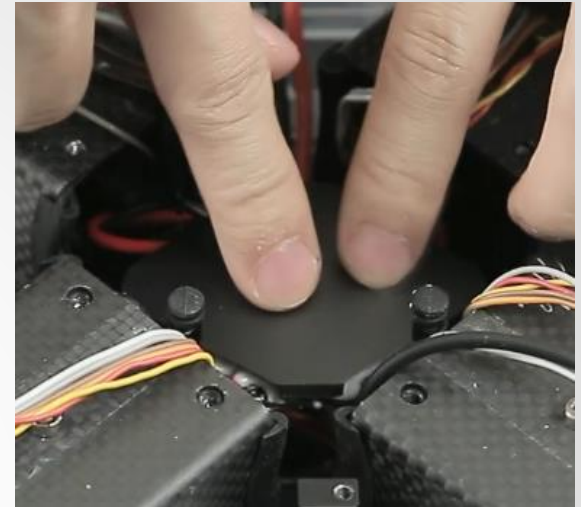
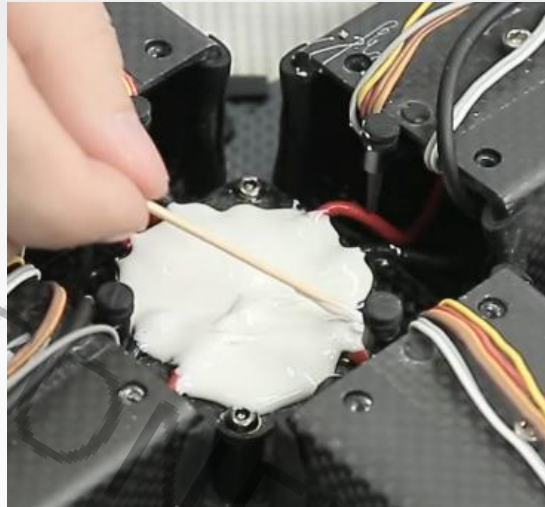
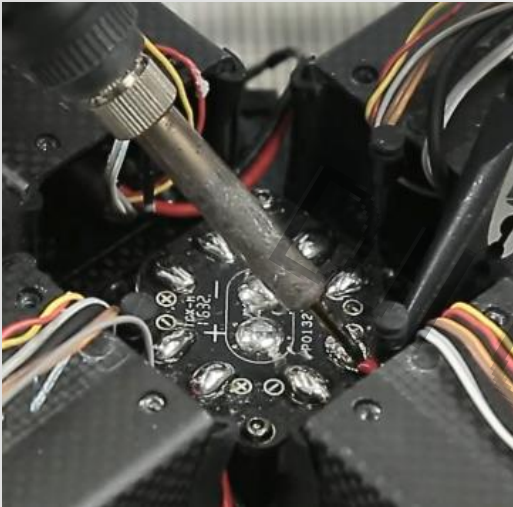
将电源线用螺丝锁紧至电调上，注意VCC对应红线，GND对应黑线，并查实电调信号线；
安装电调后盖，并锁上对应的4颗螺丝；

1. Mount the power cable to ESC and tighten it with a screw. Make sure that the red cable is welded onto ACC, black cable onto GND, and then connect the ESC signal cable.
2. Mount the rear ESC cover to the motor's base and secure the rear ESC cover with four screws.



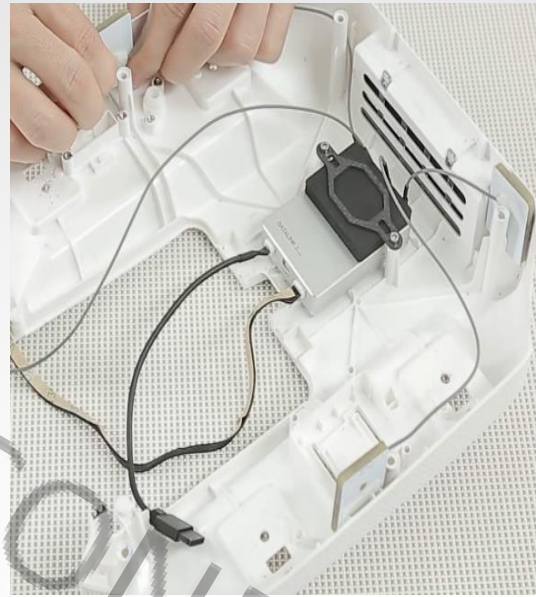
排好M1-M8号机臂，将下碳板放置在四根大臂上对准螺丝孔，
注意主控固定件位置为机头方向，
安装加强碳板，并锁入16颗螺丝，

1. Arrange M1 to M8 arms as shown. Place the lower carbon board onto the four arms and align the screw holes.
2. Make sure that the main controller mounting piece points to the aircraft's nose.
3. Mount the reinforced carbon board and tighten it with 16 screws.



将电源线焊接在电源分电板，红线焊接在“+”，黑线焊接在“-”
焊接后使用洗板水清洁焊点并涂上适量的白胶，并贴上黑色屏蔽棉；

1. Weld the power cables onto the power distribution board. Be aware that the red power cable should be welded onto the “+” position and the black cable onto the “-” position.
2. Clean the welding spots. Apply the white glue onto the power distribution board and attach a black shielding cotton.



将天空端安装在下壳并锁紧对应的固定碳片螺丝；

将3个天线粘贴至下壳侧壁上，并将下壳组件安装至机身底部，

1号天线在下壳前侧，2号天线在下壳右侧，4号天线在下壳左侧，

3号天线在中心板后侧

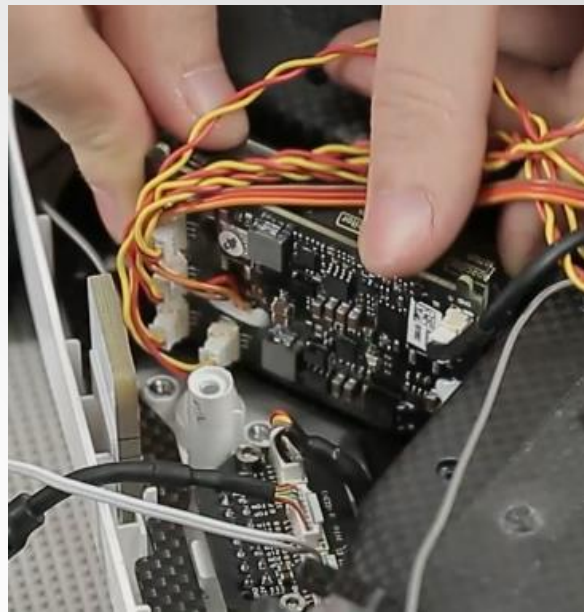
1. Mount the air system to the lower shell. Fix the carbon piece on the air system with screws.
2. Attach three antennas to the lower shell. Mount the lower shell onto the bottom of the aircraft.
3. Antenna 1 is on the front of the lower shell, antenna 2 is on the right of the lower shell, antenna 4 is on the left of the lower shell, and antenna 3 is on the rear of the central board.



将RTK接口板安装至下壳上，并锁上2颗螺丝固定；

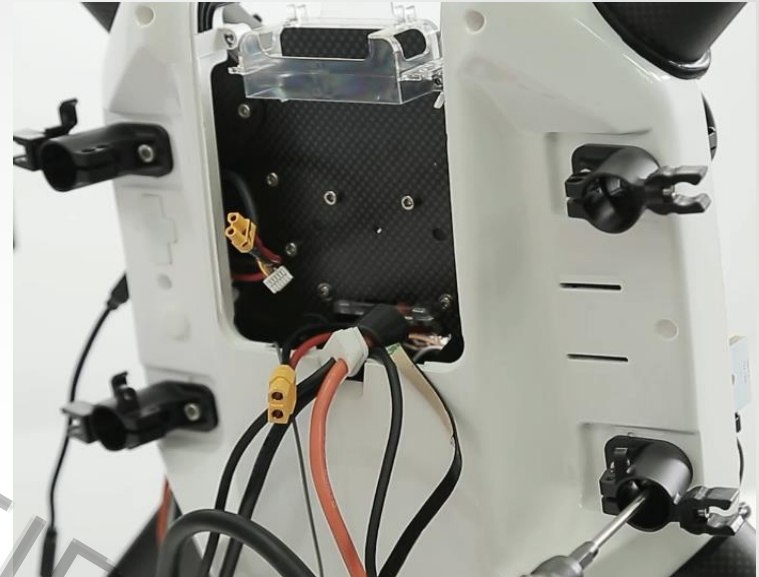
将小白壳安装在下碳板上，并锁上4颗螺丝固定；

1. Mount the RTK interface board to the lower shell and tighten it with two screws.
2. Mount the the white box on the lower carbon board , and tighten it with four screws.



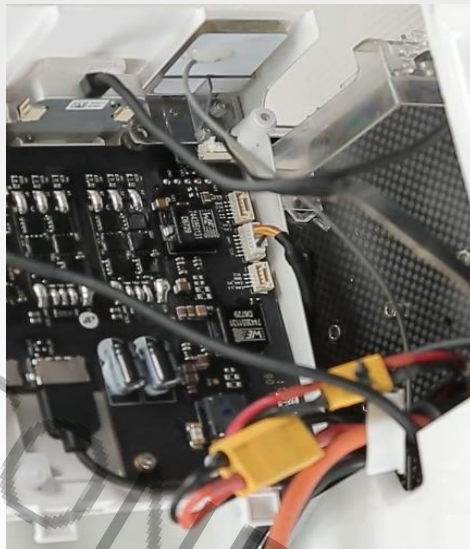
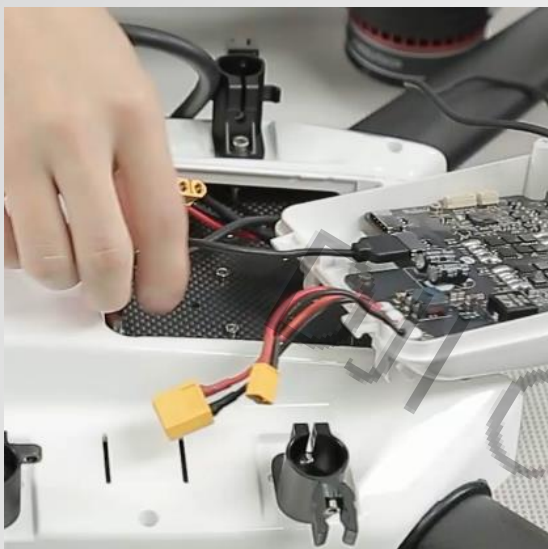
将PMU粘贴在气压计外壳上，固定PMU供电线在大臂上然后与智能插头电源线连接；
将电调馈线连接至断电续航模块上，J6接4号臂，J5接1号臂，J4接2号臂，J3接3号臂；
将断电续航模块电源线从下碳板穿出，并安装在下碳板胶钉上

1. Mount PMU to the external barometer shell. Attach the PMU power cable to the arm and connect the cable to the intelligent adapter power cable.
2. Connect the ESC cable to the reserved battery module base. Make sure the arm 4 is connected to the J6, arm 1 to J5, arm 2 to J4, and arm 3 to J3.
3. Pass the reserved battery module power cable through the lower carbon board. Mount the reserved battery module onto the carbon board.



将脚架支撑柱和碳板支撑柱固定在下壳上；注意脚架支撑柱开口需朝外；
以相同方法依次安装剩下的3个脚架支撑柱和碳板支撑柱；

1. Mount the landing gear mounting piece and carbon board mounting piece to the lower shell. Note that the connector on the landing gear mounting piece faces outwards.
2. Secure the other three landing gear mounting pieces and carbon board mounting pieces in the same way.



将中心板通讯线沿下碳板穿出；

依次将断电续航电源线、智能插头电源线、天空端信号线、RTK转接线、对频连接线连接至中心板上；

将3号天线安装在后壳侧壁卡槽内，安装后壳并锁上对应的2颗螺丝

1. Pass the central board communication cable through the lower carbon board.
2. Connect the reserved battery power cable, intelligent adapter power cable, air system signal cable, RTK adapter cable, linking cable to the central board.
3. Mount the antenna 3 to the slot on the rear shell properly. Mount the rear shell and tighten it with two screws.



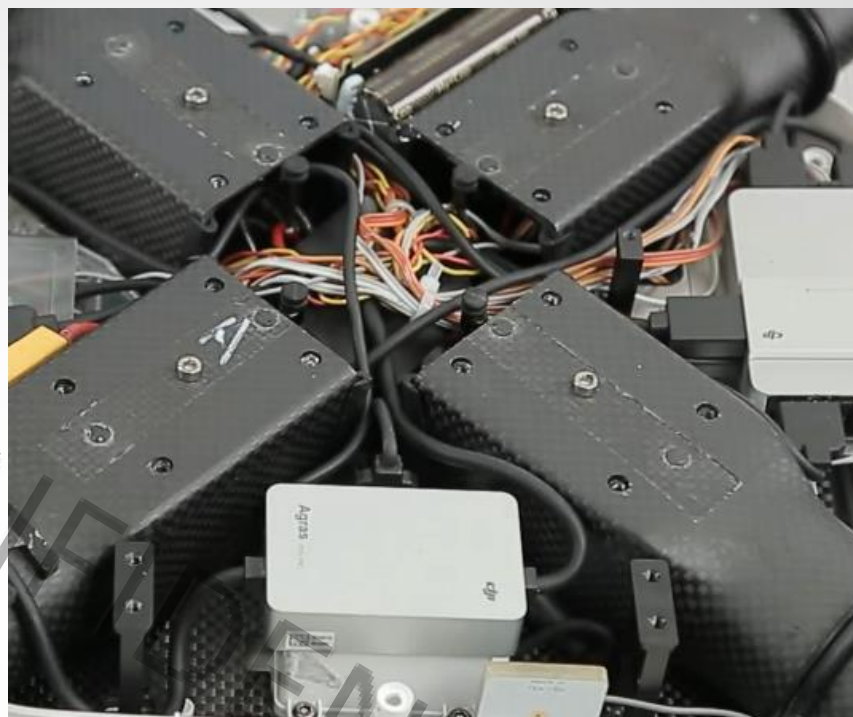
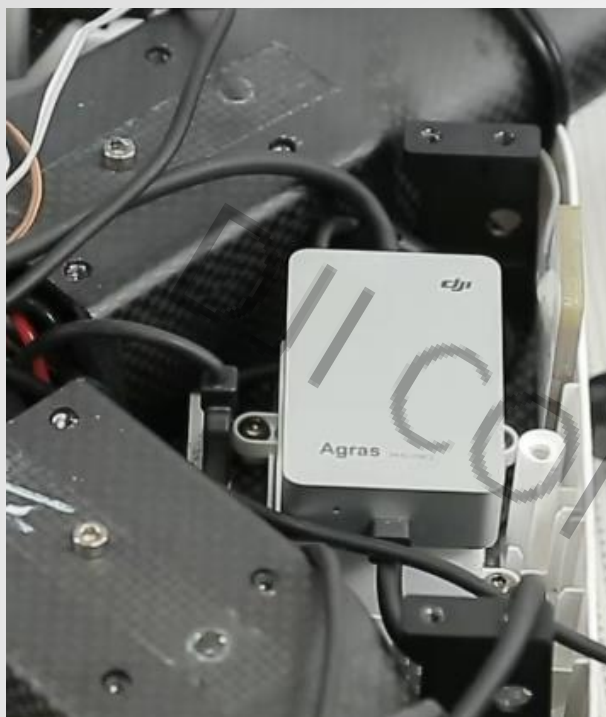
将8根电调信号线依次连接在主控M1-M8接口上，断电续航信号线连接至IESC接口；

将LED灯连接线连接到主控LED接口上，将天空端连接至RF接口；

将PMU连接线连接在主控PMU接口，RTK转接线连接至422接口；

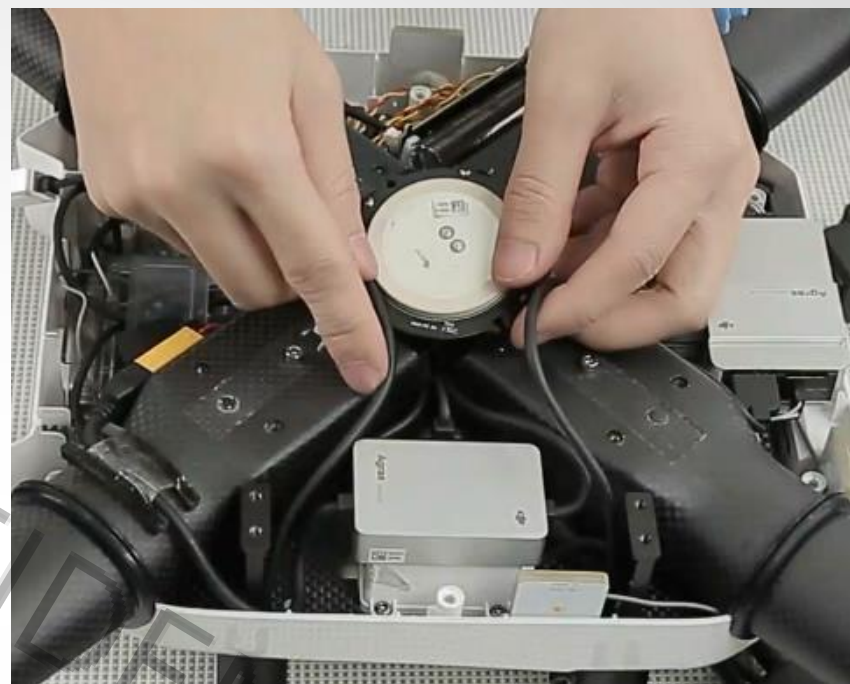
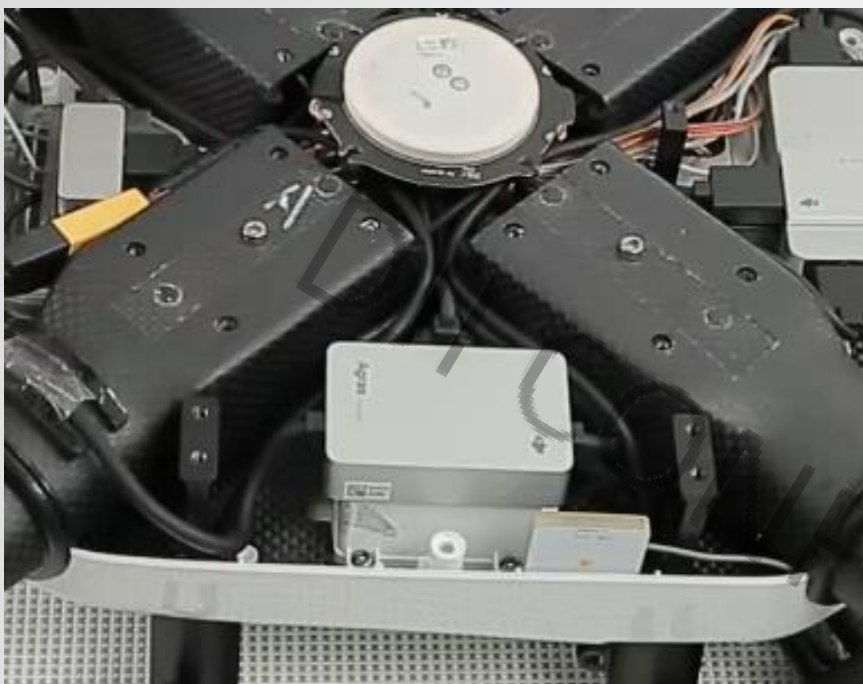
将断电续航通讯信号线和视觉转接信号线分别连接至主控CAN2接口和F8接口

1. Connect the eight ESC signal cables to the M1 to M8 ports on the main controller, the reserved battery signal cable to the ESC port.
2. Connect the LED cable to the LED port on the main controller. Connect the air system to the RF port.
3. Connect the PMU cable to the PMU port on the main controller. Connect the RTK adapter cable to the 422 port.
4. Connect the reserved battery power cable and vision signal cable to the CAN 2 port and F8 port.



将中心板通讯线连接至GPS拓展接口上；
整理线材后，并用扎带扎紧；

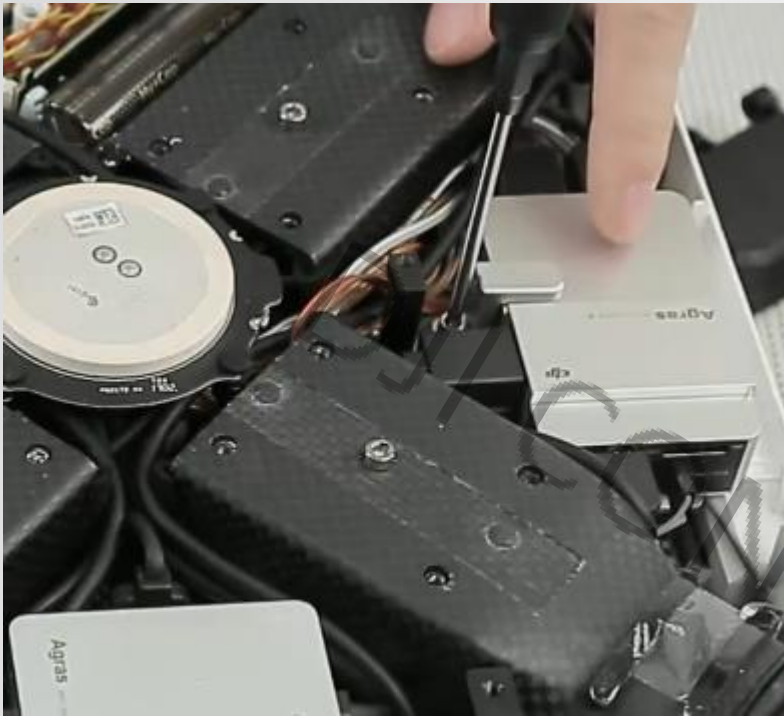
1. Connect the central board communication cable to the GPS port.
2. Arrange the cables and tighten the cable ties.



将GPS连接至主控CAN1接口，安装好GPS；

将GPS与指南针连接线沿机身右侧孔穿出，将中心板通讯线连接至GPS拓展接口上后粘贴在机尾塑料壳上；

1. Connect the GPS cable to the CAN 1 port on the main controller. Mount the GPS module.
2. Pass the cable between GPS and compass through the hole on the right side of the aircraft. Connect the Central board cable to the GPS port and then attach the plastic shell.

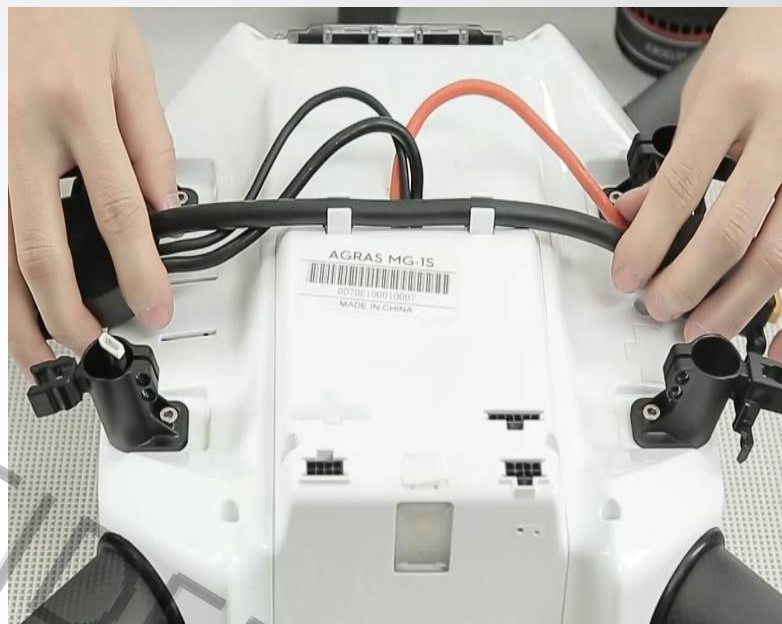


安装主控压块并锁上对应的螺丝；

安装PMU线材卡扣在上碳板并锁上1颗螺丝固定；

安装上碳板对准螺丝孔，并锁上对应的螺丝；

1. Mount the main controller mounting piece and tighten it with screws.
2. Mount the PMU cable buckle to the upper carbon board and tighten it with a screw.
Mount the upper carbon board and align with the holes. Secure the upper carbon board with screws.



安装上壳并在反面锁上对应的6颗螺丝，注意上壳需安装在橡胶槽内；
将电池电源线黑色粗线卡入槽内；

1. Mount the upper shell to the aircraft as required. Turn the aircraft to the other side and tighten the six screws.
2. Attach the black battery power cable to the slot.



将右脚架两个指南针线与GPS连接并安装线材保护套

安装脚架在支撑柱上，插入防脱片并锁入2颗螺丝固定，另一支撑柱锁入1颗利索；

以同样的方式安装左脚架；

1. Connect the two compass cables on the right landing gear to the GPS. Attach the cable cover.
2. Mount the landing gear. Attach the anti-drop piece to the landing gear and tighten it with two screws. Secure the other mounting piece with a screw.
3. Mount the left landing gear in the same way.



将电源信号插头安装至右脚架并锁入3颗螺丝固定；
用同样的方式在左脚架安装电源插头

1. Mount the power signal socket to the right landing gear and tighten it with three screws.
2. Mount the power socket to the left landing gear in the same way.

装机完毕

Assembling Completed